1. Which Cisco memory stores the startup configuration?

A) RAM
B) Flash
C) ROM
D) NVRAM

2. Which mode allows changing the router configuration?

A) User EXEC
B) Privileged EXEC
C) Global Configuration
D) ROMMON

3. The default OSPF Hello interval on Ethernet is:

A) 5 seconds
B) 10 seconds
C) 20 seconds
D) 40 seconds

4. Which router becomes DR?

A) Lowest Router ID
B) Highest Priority
C) Lowest Priority
D) Lowest Cost

5. Which protocol is connection-oriented?

A) UDP
B) IP
C) TCP
D) ARP

6. Which OSI layer performs encryption?

A) Network
B) Session
C) Presentation
D) Transport

7. Which VLAN cannot be deleted?

A) VLAN 10
B) VLAN 20
C) VLAN 100
D) VLAN 1

8. Bridge ID consists of .... and ....

A) MAC address
B) Switch ID
C) Bridge Priority
D) System ID
E) IP addresss

9. By default, all the switch ports are unprotected.

A) False
B) True

10. Switches use ... to prevent Layer 2 loops.

A) STP
B) ARP
C) VTP
D) MAC
E) ICMP

11. To form an EtherChannel, all interfaces in an EtherChannel must be configured identically.

A) True
B) False

12. Which IP address is used as the default gateway for hosts in a local network?

A) The network address
B) The broadcast address
C) The router interface IP
D) The DNS server IP

13. Which of the following IP address is a valid host in the subnet 10.0.0.0/29?

A) 10.0.0.0
B) 10.0.0.3
C) 10.0.0.7
D) 10.0.0.8

14. Which of the following subnet masks allows for 62 usable endpoints per network?

A) 255.255.255.192
B) 255.255.255.224
C) 255.255.255.248
D) 255.255.255.252

15. **For the subnet 172.16.100.128/26, Which are the valid ranges of usable IP addresses?**

A) 172.16.100.129 : 172.16.100.158
B) 172.16.100.130 : 172.16.100.161
C) 172.16.100.128 : 172.16.100.190
D) 172.16.100.131 : 172.16.100.190

16. PC can't access the internet, but it can ping other devices in the same network, What is the most likely cause?

A) Incorrect subnet mask
B) Missing default gateway
C) Wrong MAC address
D) DNS server failure

17. A host with IP address 192.168.10.25/24 wants to communicate with 192.168.20.10. What will the host do first?

A) Send the packet directly using ARP
B) Forward the packet to the default gateway
C) Broadcast to find the destination IP
D) Drop the packet Ans: Forward the packet to the default gateway

18. A router has 2 interfaces: 192.168.1.1/24 and 172.16.0.1/16. Which network cannot reach it directly?

A) 192.168.1.50
B) 172.16.5.0
C) 192.168.0.50
D) 172.16.255.200

19. A router loses power before the administrator saves the configuration. Which data will still be available after reboot?

A) Running configuration only
B) Startup configuration only
C) Running configuration and routing table
D) ARP cache only

20. Which protocol does NOT guarantee delivery?

A) TCP
B) FTP
C) UDP
D) HTTPS

21. Which command limits failed SSH login attempts to three?

A) ip ssh timeout 3
B) ip ssh authentication-retries 3
C) login retries 3
D) ssh retries 3

22.An administrator configured SSH but forgot to configure login local on the VTY lines.

What is the most likely result?

A) SSH works normally.
B) The router will use the local username database automatically.
C) User authentication will fail because the VTY lines are not configured to use the local database.
D) SSH falls back to Telnet.

23.If two OSPF routers have different Hello timers, what happens?

A) They become neighbors.
B) They exchange routes normally.
C) Adjacency will NOT form.
D) DR election still happens.

24.A switch receives a frame with an unknown destination MAC address.

What happens?

A) Drops the frame
B) Sends to router
C) Floods all ports except the incoming port
D) Sends to gateway

25.What is the main advantage of dynamic routing over static routing?

A) It always provides better security.
B) It automatically adapts to topology changes without manual intervention.
C) It requires less CPU and memory.
D) It does not exchange routing information.